

Hedging — Pathwise Delta / Gamma and a Daily-Rebalanced Hedge

Risk sensitivities and a **daily-rebalanced delta hedge** for the swing option at the focal regime ($c = 0.04$, $\gamma = 2$), comparing three pricers throughout: **LSM** (full-state Least-Squares Monte Carlo benchmark), **RL** (D4PG trained with the single-sample TD bootstrap, *no kernel*), and **RL-kernel** (D4PG trained with the semi-analytical expected backup over the HHK transition kernel).

Greeks — CRN bump-and-revalue. $V(S_0)$ is revalued under the frozen policy on path bundles started at bumped initial spots sharing one simulation seed; in HHK the seed-keyed draws are independent of S_0 , so the finite-difference noise cancels path-by-path. See `src/greeks.py`, `tools/test_greeks.py`.

Daily hedge. At each decision date the per-path continuation Delta $\Delta_t = \partial V_t / \partial S_t$ is obtained by bumping the prompt spot and re-rolling the policy to maturity (the bump propagates through the OU factor in closed form, $X_k \rightarrow X_k + \delta e^{-\alpha(t_k - t)}$). The pathwise delta anticipates the future, so it is projected onto the date- t state (Longstaff–Schwartz) to make the hedge ratio $\mathcal{F}_t$ -measurable. We hedge with the HHK forward to maturity; the discounted P&L increment per contract is the martingale difference $DF_{t+1}(F_{t+1} - F_t)$.

Switching kernel / no-kernel. The two RL variants are trained with **different canonical configurations** (the kernel makes the TD target deterministic, so critic LR, exploration schedule, depth, and gradient steps differ; both flag sets live in `tools/train_hedging_agents.py`). Each entry in the `AGENTS` registry below is rebuilt from its own saved-run JSON, so adding, removing, or swapping a variant is a one-line change.

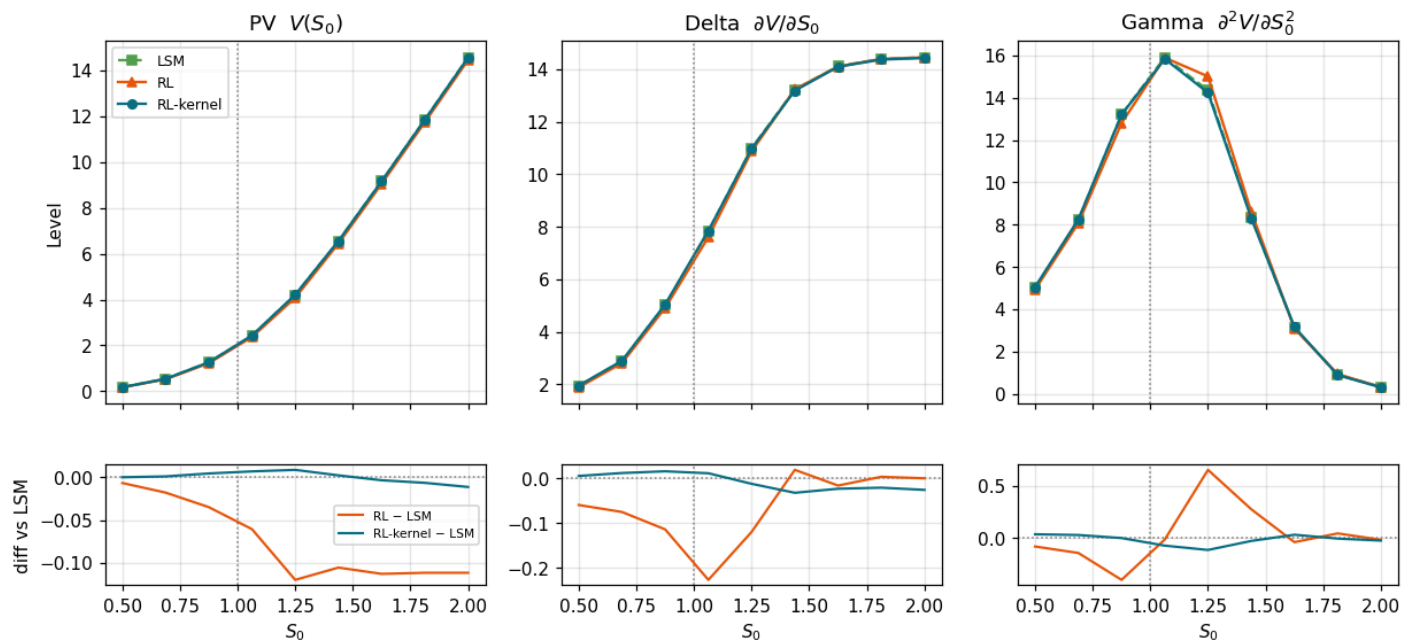
Contract: $S_0=1.0$, $K=1.0$, $n_rights=22$, $c=0.04$, $\gamma=2.0$ | HHK $\alpha=12.0$

AGENTS registry — each variant is rebuilt from its saved-run JSON

variant	run	kernel	episodes	layers	squash	lr_c	noise	eval EMA
RL-kernel	v64_32k_11	✓	32,768	3	beta_sigmoid_1 .5	0.0006	linear	ema
RL	nokernel_32k_1 1	×	32,768	2	beta_sigmoid_ 3.0	0.0003	hyperbolic	off

1. PV, Delta, Gamma over a 9-point S_0 grid — revalued

Grid spans $[0.5 S_0, 2.0 S_0]$. PV is **revalued** at each node — RL via CRN policy revaluation, LSM by refitting and pricing out-of-sample at that spot — and Delta/Gamma are central finite differences of the revalued PV grid. All three methods share the test seed.

Revalued risk profile on a 9-pt grid $[0.5, 2.0] \cdot S_0$ — LSM / RL / RL-kernel

PV, Delta ($\partial V / \partial S_0$), Gamma ($\partial^2 V / \partial S_0^2$) on the 9-point revaluation grid. $\Delta PV = RL - LSM$ (green = above, red = below).

	S_0	PV LSM	PV RL	PV RL-kernel	ΔPV RL	ΔPV RL-kernel	Δ LSM	Δ RL	Δ RL-kernel	Γ LSM	Γ RL	Γ RL-kernel
0	0.500	0.179	0.172	0.179	-0.006	+0.000	1.942	1.883	1.948	5.021	4.938	5.056
1	0.688	0.543	0.525	0.544	-0.018	+0.001	2.884	2.809	2.896	8.221	8.076	8.249
2	0.875	1.260	1.226	1.265	-0.035	+0.005	5.025	4.911	5.041	13.214	12.809	13.212
3	1.062	2.428	2.367	2.435	-0.061	+0.007	7.839	7.612	7.850	15.916	15.901	15.842
4	1.250	4.200	4.080	4.209	-0.120	+0.009	10.994	10.874	10.982	14.358	15.014	14.241
5	1.438	6.550	6.445	6.553	-0.106	+0.002	13.223	13.242	13.191	8.342	8.619	8.313
6	1.625	9.159	9.046	9.155	-0.113	-0.003	14.122	14.106	14.099	3.126	3.085	3.157
7	1.812	11.846	11.735	11.840	-0.112	-0.006	14.396	14.399	14.375	0.894	0.937	0.887
8	2.000	14.557	14.445	14.546	-0.112	-0.011	14.457	14.457	14.432	0.330	0.312	0.304

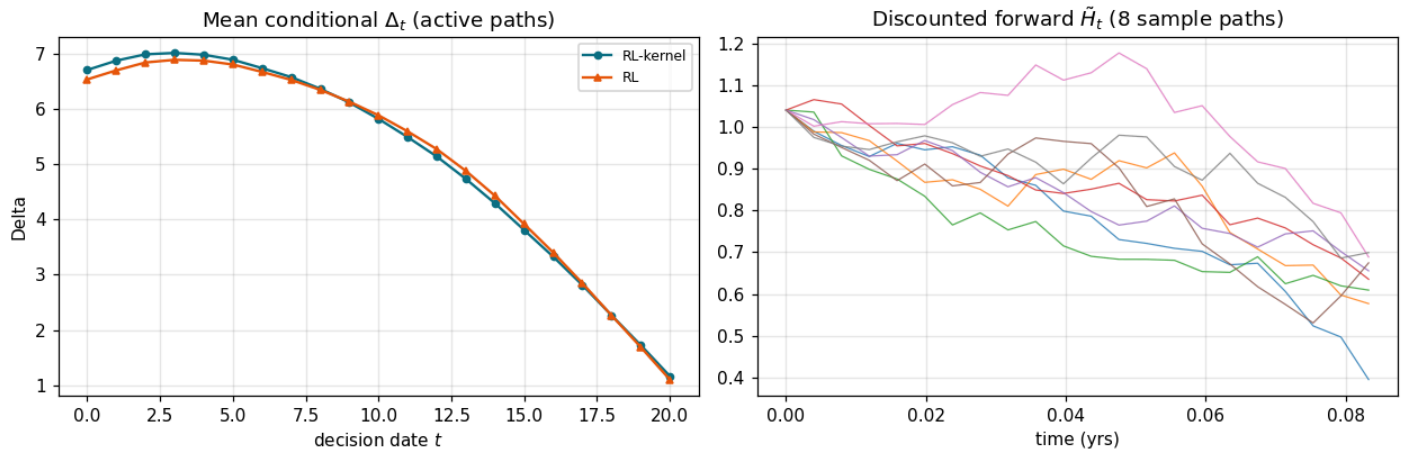
RL-kernel reproduces the LSM value function to within ± 0.01 PV units ($\lesssim 0.5\%$) over the entire $[0.5, 2.0] \cdot S_0$ range, with Delta and Gamma profiles that coincide with the benchmark. **RL (no kernel)** prices systematically below both — the shortfall is largest near the money ($\approx -2.5\%$ at $S_0 \approx 1.06$) — yet its Greeks stay close to the common profile: the no-kernel deficit is an exercise-quality effect on value, not a different risk profile.

2. Daily-rebalanced delta hedge — both RL policies

Per-date continuation Delta by spot-bump-and-re-roll, conditioned on the date- t state (Longstaff-Schwartz projection), hedged with the HHK forward to maturity. The same procedure runs on each RL variant's own policy and cashflows.

RL-kernel PV=1.9819 | unhedged std 2.5963 → hedged std 1.4025 (mean +0.0050; var reduction 70.8%)

RL PV=1.9221 | unhedged std 2.6039 → hedged std 1.4141 (mean +0.0073; var reduction 70.5%)



The hedged P&L mean is ≈ 0 for both policies — the forward is a fair martingale and the hedge ratio is $\mathcal{F}_t$ -measurable, so the hedge is non-anticipating — while the variance drops by $\approx 70\%$. Hedge quality is essentially identical across the two RL variants: the kernel's contribution is pricing accuracy, not hedgeability.

3. A common hedge for all three pricers — regression delta

To compare RL and LSM on equal footing we hedge **all three** with the same **regression (Longstaff–Schwartz) delta**: at each date the realised continuation value is regressed on the date- t state and differentiated in spot. This single $\mathcal{F}_t$ -measurable estimator runs identically on RL and LSM cashflows. For the RL policies we additionally keep the refined continuation-bump hedge from §2, which the differentiable policies make possible.

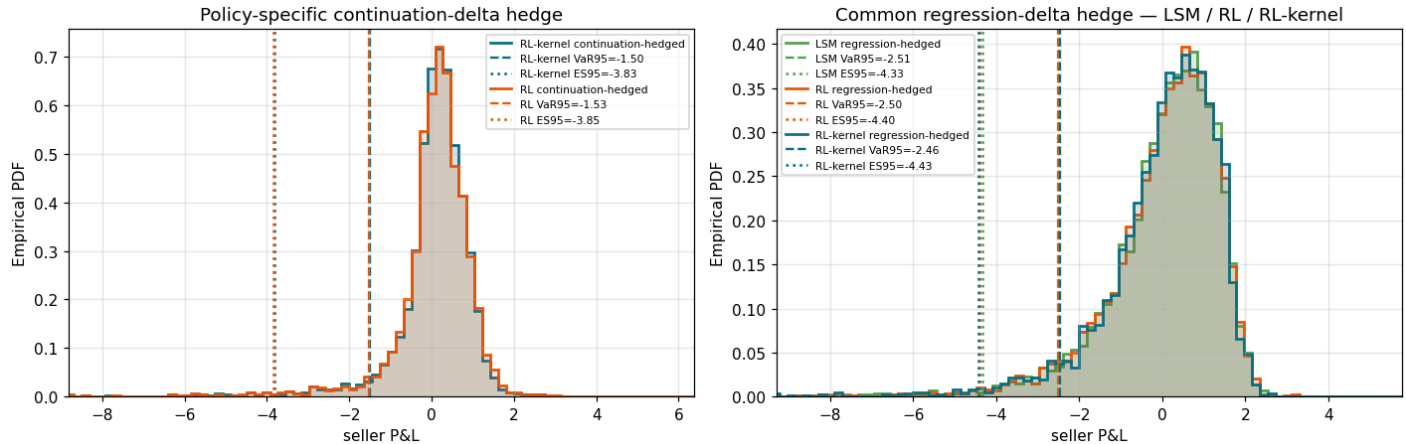
LSM price (scalar) 1.9828 vs per-path pv mean 1.9828

Seller P&L risk metrics — lower std / more negative ES = more risk; higher std-reduction % = better hedge.

	mean_pnl	variance_pnl	std_pnl	var_pnl_95	cvar_pnl_95	min_pnl	max_pnl	std reduction %
LSM unhedged	-0.000000	6.176500	2.485300	-5.093200	-7.013900	-18.991100	1.982800	—
LSM regression-hedged	0.005400	2.200900	1.483500	-2.509000	-4.333600	-18.780500	4.677800	40.3
RL unhedged	0.000000	6.780400	2.603900	-5.059600	-7.212400	-51.692700	1.922100	—
RL regression-hedged	0.002100	2.821800	1.679800	-2.499700	-4.404800	-50.084800	3.182100	35.5
RL-kernel unhedged	-0.000000	6.740600	2.596300	-4.995900	-7.159400	-51.727600	1.981900	—
RL-kernel regression-hedged	-0.001300	2.788200	1.669800	-2.456900	-4.431600	-49.967300	2.701500	35.7
RL-kernel continuation-hedged	0.005000	1.967000	1.402500	-1.504000	-3.828900	-49.755400	5.086100	46.0
RL continuation-hedged	0.007300	1.999700	1.414100	-1.534000	-3.852700	-49.795100	5.282400	45.7

4. Final P&L distributions with VaR & ES

95% **VaR** (5th percentile of seller P&L) and **ES** (mean of the 5% left tail) as vertical lines. Left: the policy-specific continuation-bump hedge (RL vs RL-kernel). Right: the common regression-delta hedge across all three pricers.



5. Takeaways

- **Pricing across spot (§1).** RL-kernel matches the LSM benchmark over the full $[0.5, 2.0] \cdot S_0$ grid (PV within ± 0.01 ; Delta/Gamma coincide). RL without the kernel sits below both everywhere, worst near the money ($\approx -2.5\%$) — consistent with the pricing gap quantified in Validation 3.
- **The daily hedge works (§2).** The conditioned continuation-Delta forward hedge yields ≈ 0 -mean seller P&L with $\approx 70\%$ variance reduction for both RL policies; the hedge ratio is $\mathcal{F}_t$ -measurable.
- **A common yardstick (§3–4).** Under the identical regression-delta hedge, all three pricers cut P&L std by 35–40% with nearly identical VaR/ES — hedge effectiveness does not separate them. The continuation-bump hedge, available only for the differentiable RL policies, does better: $\approx 46\%$ std reduction and the tightest tail (ES95 ≈ -3.8 vs $-4.3/-4.4$).
- **Where each method earns its place.** The kernel buys value-function accuracy (LSM-parity pricing); the differentiable policy buys an exact per-state continuation delta that the LSM capacity-DP cannot provide (it falls back on the regression delta). A DP-consistent LSM delta surface is a natural extension.